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Statistical Quality Control

Seminar Paper

Simultaneous (Joined) Control Charts

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Simultaneous (Joined) $\bar{X}$ and R/S Control Charts for Effective Process Monitoring

1. Introduction and General Discussion

The fundamental theoretical framework and methodological approach for this seminar paper are largely derived from **Douglas C. Montgomery's seminal work, 'Introduction to Statistical Quality Control.'** Montgomery's insightful analyses and comprehensive treatment of Statistical Process Control have been instrumental in guiding our understanding of the core principles, effectiveness, and critical practical importance of using $\bar{X}$ and R control charts simultaneously. Throughout this study, both the theoretical insights and the practical applications presented draw significantly from the robust statistical foundations emphasized by Montgomery.

Quality is crucial for any product or service. When customers buy something, they expect it to work well and be consistent. **Statistical Process Control (SPC)** is a powerful set of tools that helps us make sure our production processes are stable and produce high-quality goods consistently. At the heart of SPC are **control charts**, which are special graphs that show us if a process is running as it should.

Every process naturally has some **variation**. Think of making cookies: no two cookies will be exactly the same size. This expected, minor variation comes from **common causes** – the normal part of the process. We can't easily remove these. However, sometimes there's unexpected, larger variation due to **assignable causes**. These are problems, like a broken machine part or a new operator who isn't fully trained. Control charts are designed to help us quickly spot these assignable causes so we can fix them.

A manufacturing process can be described by two main features: its **average** (or mean) and its **spread** (or variation). The average tells us about the central value of a product characteristic, for example, the target weight of a package. The variation tells us how consistent that characteristic is, or how much individual packages differ in weight.

It is **critically important** to monitor **both** the mean and the variation of a process. Why? Because looking at only one can be misleading. As Montgomery emphasizes, "It is very important to use control charts for both the process mean and process variability **together**" Imagine a juice bottling machine:

- If we only check the **average** volume, bottles might seem fine, but some could be almost empty while others overflow. This shows high variation even if the average is correct.
- If we only check the **variation** (how consistent the fill is), bottles might be filled to very similar levels, but consistently too low. The mean is off, even if variation is low.

These examples clearly show why **joint monitoring** of the mean and variation is essential. This paper will explain how **Simultaneous Joined $\bar{X}$ and R/S Charts** allow us to to do this effectively. We will show these charts and how to use them with real data.

It's also important to understand that this approach is different from **multivariate control charts**. In our case, we're monitoring **one single product characteristic** (e.g., length) using **two separate univariate (single-variable) control charts** ($\bar{X}$ for the mean and R or S for variation). Multivariate charts, on the other hand, are used when we want to monitor **two or more different product characteristics at the same time** (e.g., a product's length, width, and weight) and understand how they interact. Our focus is on getting a complete picture of one characteristic's performance.

2. Theoretical Part: Understanding $\bar{X}$ and R/S Control Charts

To monitor a process, we collect data in small groups called **subgroups**. Each subgroup contains n individual measurements taken at a specific time. We then collect many such subgroups, let's say m subgroups, over time.

2.1 Control Charts for Process Variability: The R and S Charts

Before checking the process average, we **must always check the process variability first**. This is a fundamental rule in SPC. **Why?**

Because the control limits for the $\bar{X}$ (mean) chart depend on how stable the process variation is. If the variation is out of control, the limits on the $\bar{X}$ chart will not be reliable or meaningful.

2.1.1 The R Chart (Range Chart)

The R chart is used to monitor process variation when the subgroup size (n) is small, typically $n \leq 10$.

- ***Range (Ri):***

For each subgroup i, the range is simply the difference between the largest value and the smallest value.

$$R_i = X_{max,i} - X_{min,i}$$

Where:

- Xmax,i is the largest observation in subgroup i.
- Xmin,i is the smallest observation in subgroup i

- ***Average Range ($\bar{R}$):***

  This is the average of all the individual subgroup ranges. It forms the center line (CL) for our R chart.

$$\bar{R} = \frac{R_1 + R_2 + \cdots + R_m}{m}$$

  Where:

  - Ri is the range of subgroup i.
  - m is the total number of subgroups.

- ***Control Limits for R Chart:***

  These define the boundaries of expected variation.

  - Center Line (CL). : CLR = $\bar{R}$
  - Upper Control Limit (UCL) : UCLR = $D_4\bar{R}$
  - Lower Control Limit (LCL) : LCLR = $D_3\ \bar{R}$

  Where:

  - $D_3$and D_4 are special control chart constants. Their values depend on the subgroup size (n). These constants are found in standard tables, such as Table VI in D.C. Montgomery's Introduction to Statistical Quality Control Book (Appendix VI Factors for Construcuting Variables Control Charts, p.A-12, 8th edition). If the calculated LCLR is negative, we use 0 instead, as range cannot be negative.
  - For n=5, Montgomery's Table VI gives D_3=0 and D_4=2.114.
  - These assume a false alarm rate (α) of 0.27% per chart (3-sigma).

2.1.2 The S Chart (Standard Deviation Chart)

The S chart is preferred for monitoring process variation when the subgroup size (n) is larger, typically n>10. It provides a more accurate estimate of the process standard deviation than the range.

- ***Sample Standard Deviation (Si):***

  For each subgroup i, the standard deviation measures the spread of data around its mean.

- ***Sample Standard Deviation (Si):***

  For each subgroup i, the standard deviation measures the spread of data around its mean

$$S_i = \sqrt{\frac{\sum_{j=1}^{n}(X_{ij} - \bar{X}_i)^2}{n-1}}$$

  Where:

  - Xij is the j-th observation in subgroup i.
  - $\overline{X}$ i is the mean of subgroup i.
  - n is the subgroup size.
  - Si represents the sample standard deviation for a specific subgroup **i**. It quantifies the typical amount of variation or dispersion of the individual data points (Xij) around their own subgroup mean ($\overline{X}$ i) within that particular sample. A larger Si indicates more spread-out data, while a smaller Si indicates data points are clustered more closely around the mean..

- ***Average Standard Deviation ($\bar{S}$):***

  This is the average of all the individual subgroup standard deviations. It forms the center line (CL) for our S chart.

$$\bar{S} = \frac{S_1 + S_2 + \cdots + S_m}{m}$$

  or, using summation notation:

$$\bar{S} = \frac{\sum_{i=1}^{m} S_i}{m}$$

  Where:

  - **$\bar{S}$ (S-bar)** represents the **Average Standard Deviation**. This is the central line (CL) for our S-chart. We calculate it by taking all the individual standard deviations from each subgroup and finding their average. It provides an overall estimate of the process variability across all the collected subgroups.

  - **Si (S sub i)** refers to the Sample Standard Deviation for an individual subgroup **i.** We calculate this value for each subgroup we collect, as we discussed previously. It measures the spread of data within that specific subgroup.
  - **m** represents the total number of subgroups (or samples) you have collected. When you calculate $\overline{S}$, you are summing up the Si values for all m subgroups and then dividing by m to get the average.
  -

- ***Control Limits for S Chart:***

  - Center Line (CL): CLs = $\bar{S}$
  - Upper Control Limit (UCL): UCLs = $B_4\bar{S}$
  - Lower Control Limit (LCL): LCLs = $B_3\bar{S}$

  Where:

  - B_3 and B_4 are control chart constants that depend on the subgroup size (n). These are also found in standard tables (e.g., Table VI in Montgomery, Appendix A). If the calculated LCLS is negative, we use 0.
  - B_3 and B_4 for n=5, Montgomery's Table VI gives B_3=0 and B_4 2.089.
  - These assume a false alarm rate (α) of 0.27% per chart (3-sigma).

2.2 Control Chart for Process Mean: The $\bar{X}$ (X-bar) Chart

The $\bar{X}$ chart is used to monitor the process average. As mentioned, it should only be interpreted after the R or S chart shows the process variability is in statistical control.

- ***Sample Mean ($\bar{X}i$):***

  For each subgroup i, this is simply the average of all observations in that subgroup.

$$\bar{X}_i = \frac{X_{i1} + X_{i2} + \cdots + X_{in}}{n}$$

  Where:

  - Xij is the j-th observation in subgroup i.
  - n is the subgroup size.

- ***Grand Average ($\bar{\bar{X}}$):***

  This is the average of all the individual subgroup means. It forms the center line (CL) for our $\bar{X}$ chart.

$$\bar{\bar{X}} = \frac{\bar{X}_1 + \bar{X}_2 + \cdots + \bar{X}_m}{m}$$

  Or, using summation notation:

$$\bar{\bar{X}} = \frac{\sum_{i=1}^{m} \bar{X}_i}{m}$$

Where:

- $\bar{X}i$ is the j-th observation in subgroup i.
- m represents the total number of subgroups (or samples) collected.
- Means The Grand Average ($\bar{\bar{X}}$) is calculated by summing up the average of each subgroup ($\bar{X}i$) and then dividing by the total number of subgroups. If you have collected data from 20 different subgroups, then m would be 20.

- ***Control Limits for $\bar{X}$ Chart (Using $\bar{R}$ from R Chart):***

  - Center Line (CL). : $CL_{\bar{x}} = \bar{\bar{X}}$
  - Upper Control Limit (UCL) : $UCL_{\bar{x}} = \bar{\bar{X}} + A_2\bar{R}$
  - Lower Control Limit (LCL) : $LCL_{\bar{x}} = \bar{\bar{X}} - A_2\bar{R}$

  Where:

  - A_2 is a control chart constant that depends on the subgroup size (n) and is found in the same tables as D_3,D_4
  - A_2 is a control chart constant. For n=5, Montgomery's Table VI gives A_2 =0.577
  - These assume a false alarm rate (α) of 0.27% per chart (3-sigma).

- ***Control Limits for $\bar{X}$ Chart (Using $\bar{S}$ from S Chart)***

  - Center Line (CL). : $CL_{\bar{x}} = \bar{\bar{\bar{X}}}$
  - Upper Control Limit (UCL): $UCL_{\bar{x}} = \bar{\bar{\bar{X}}} + A_3\bar{S}$
  - Lower Control Limit (LCL): $LCL_{\bar{x}} = \bar{\bar{X}} - A_3\bar{S}$

  Where:

  - A_3 is a control chart constant that depends on the subgroup size (n) and is found in the same table as above explained in B_3,B_4
  - $A_3$is a control chart constant. For n=5, Montgomery's Table VI gives $A_3$1.427
  - These assume a false alarm rate (α) of 0.27% per chart (3-sigma).

Why $\bar{\mathbf{X}}$ Chart Limits Depend on $\bar{\mathbf{R}}$ or $\bar{\mathbf{S}}$ Charts ?

- The $\bar{\mathbf{X}}$ chart checks if your process average is stable. It tells us our product's typical size, weight, or other measurement is staying consistent.
- The $\bar{R}$ chart (or $\bar{\mathbf{S}}$ chart) checks if our process variability is stable. This means it tells us if our products are consistently similar to each other. Are they all about the same size, or is there a lot of difference between them?

Here's the key connection:

The control limits for our $\bar{X}$ chart are based on your process's natural variability. This means how much your process normally changes from one item to the next.

- If our process variability (shown by $\bar{R}$ or $\bar{S}$) increases, the $\bar{X}$ chart's limits will get wider. This is because the "normal" amount of change for our average has grown.
- If our process variability decreases, the $\bar{X}$ chart's limits will get narrower. This is because our average is now expected to vary less.

3. Joint Evaluation: Interpreting $\bar{X}$ and R/S Charts Together

This is the core of "**simultaneous control charts**" in the context of Shewhart charts. We use these two separate charts to get a complete picture of process stability.

3.1 The Importance of Sequential Analysis (from Montgomery's Perspective)

As highlighted by Montgomery (2021), a strict order should be followed when analyzing these charts:

1. **Always analyze the variability chart (R or S chart) first.**

   If the R or S chart shows the process variability is **out of control** (e.g., points outside limits, non-random patterns), then the process is unstable. The calculated control limits for the $\bar{X}$ chart are not correct because they are based on this unstable variability. In this situation, we must stop, find the assignable causes affecting the variability (e.g., worn tools, unstable raw material), eliminate them, and then recalculate the control limits for both charts with the corrected data.

2. **Only if the variability chart is in control, then analyze the mean chart ($\bar{X}$ chart).**

   If the R or S chart shows that the process variability is stable (in control), then we can be confident that the control limits for the $\bar{X}$ chart are valid. We can then look for out-of-control signals on the $\bar{X}$ chart. If the variability is not under control (meaning the $\bar{R}$ or $\bar{S}$ chart shows problems), then the limits for our $\bar{X}$ chart won't be reliable.We need a stable amount of variation to correctly understand if our average is stable.

3.2 Interpreting "Out-of-Control" Signals

A process is considered "**out of control**" if points on the chart show unusual behavior.

Common signals include:

- *Points outside the control limits (UCL or LCL):* This is the most obvious sign. A point above the UCL or below the LCL suggests a sudden, significant change in the process.

- *Non-random patterns (Run Rules):*

  Even if points are within limits, certain patterns can indicate a problem. Examples include:

  - **Runs** : A sequence of several points (e.g., 7 or more) all above or all below the center line, suggesting a shift in the mean.
  - **Trends** : A steady increase or decrease of several points, showing a gradual change in the process.
  - **Cycles :** Repeated patterns of high and low points, which might indicate environmental factors or machine wear. These patterns help detect more subtle shifts that a single out-of-limit point might miss.

3.4 Advantages of Simultaneous (Joint) $\bar{X}$ and R/S Charts

Using $\bar{X}$ and R/S charts together provides significant benefits for quality control:

- **Complete Process Picture:**
  We get a full, two-dimensional view of process performance. This shows us if both the average and consistency are stable.
- **Early Problem Detection:**
  These charts help us find assignable causes quickly. It does not matter if these causes affect the process mean (like a wrong machine setting) or the process variation (like worn equipment).
- **Clear Guidance for Action:**
  When an out-of-control signal appears, the charts help us understand what kind of problem it is. This guides us to the correct root cause. For example, a problem on the R chart points to issues with consistency, while a problem on the $\bar{X}$ chart suggests a shift in the average.
- **Foundation for Improvement:**
  By regularly monitoring and controlling processes, organizations can identify areas for improvement. This helps to reduce waste and improve customer satisfaction.

4. Practical Part: Analysis of Flow Width Measurements

In this section, we will apply the principles we discussed to a real dataset. Shown in below as Dataset I and Dataset II) .This data comes from Montgomery's textbook (8th Ed., pp. 223 & 227). It shows **"Flow width measurements (in microns) for a hard bake process used for wafers"**. We have 45 subgroups, and each subgroup contains 5 measurements.

First, we will analyze the data as it is given in the book. This usually means the process is "**in control.**" We call this **Phase I analysis**. Our goal in Phase I is to check if the process was stable in the past and to set up correct control limits. After this, we will change the data a little bit. We will create "out-of-control" situations. This will help us see how the charts react and what we should do when problems happen.

Data Set I

Sample Number	1	2	3	4	5	$\bar{X}_i$	Ri
1	1,3235	1,4128	1,6744	1,4573	1,6914	1,5119	0.3679
2	1,4314	1,3592	1,6075	1,4666	1,6109	1,4951	0.2517
3	1,4284	1,4871	1,4932	1,4324	1,5674	1,4817	0.1390
4	1,5028	1,6362	1,3841	1,2831	1,5507	1,4712	0.3521
5	1,5604	1,2735	1,5265	1,4363	1,6441	1,4882	0.3706
6	1,5955	1,5451	1,3574	1,3281	1,4198	1,4492	0.2674
7	1,6274	1,5064	1,8366	1,4177	1,5144	1,5805	0.4189
8	1,4190	1,4303	1,6637	1,6067	1,5519	1,5343	0.2447
9	1,3884	1,7277	1,5355	1,5176	1,3688	1,5076	0.3589
10	1,4039	1,6697	1,5089	1,4627	1,5220	1,5134	0.2658
11	1,4158	1,7667	1,4278	1,5928	1,4181	1,5242	0.3509
12	1,5821	1,3355	1,5777	1,3908	1,7559	1,5284	0.4204
13	1,2856	1,4106	1,4447	1,6398	1,1928	1,3947	0.4470
14	1,4951	1,4036	1,5893	1,6458	1,4969	1,5261	0.2422
15	1,3589	1,2863	1,5996	1,2497	1,5471	1,4083	0.3499
16	1,5747	1,5301	1,5171	1,1839	1,8662	1,5344	0.6823
17	1,3680	1,7269	1,3957	1,5014	1,4449	1,4874	0.3589
18	1,4163	1,3864	1,3057	1,6210	1,5573	1,4573	0.3153
19	1,5796	1,4185	1,6541	1,5116	1,7247	1,5777	0.3062
20	1,7106	1,4412	1,2361	1,3820	1,7601	1,5060	0.5240
21	1,4371	1,5051	1,3485	1,5670	1,4880	1,4691	0.2185
22	1,4738	1,5936	1,6583	1,4973	1,4720	1,5390	0.1863
23	1,5917	1,4333	1,5551	1,5295	1,6866	1,5592	0.2533
24	1,6399	1,5243	1,5705	1,5563	1,5530	1,5688	0.1156
25	1,5797	1,3663	1,6240	1,3732	1,6887	1,5264	0.3224

Data set II

Sample Number	1	2	3	4	5	$\bar{X}_i$	Ri
26	1,4483	1,5458	1,4558	1,4303	1,6206	1,4998	0,1903
27	1,5435	1,6899	1,5830	1,3358	1,4187	1,5142	0,3541
28	1,5175	1,3446	1,4723	1,6657	1,6661	1,5332	0,3215
29	1,5454	1,0931	1,4072	1,5039	1,5264	1,4152	0,4523
30	1,4418	1,5059	1,5124	1,4620	1,6263	1,5097	0,1845
31	1,4301	1,2725	1,5945	1,5397	1,5252	1,4724	0,3220
32	1,4981	1,4506	1,6174	1,5837	1,4962	1,5292	0,1668
33	1,3009	1,5060	1,6231	1,5831	1,6454	1,5317	0,3445
34	1,4132	1,4603	1,5808	1,7111	1,7313	1,5793	0,3181
35	1,3817	1,3135	1,4953	1,4894	1,4596	1,4279	0,1818
36	1,5765	1,7014	1,4026	1,2773	1,4541	1,4824	0,4241
37	1,4936	1,4372	1,5139	1,4808	1,5293	1,4910	0,0920
38	1,5729	1,6738	1,5048	1,5651	1,7473	1,6128	0,0243
39	1,8089	1,5513	1,8250	1,4389	1,6558	1,6560	0,3861
40	1,6236	1,5393	1,6738	1,8698,	1,5036	1,6420	0,3662
41	1,4120	1,7931	1,7345	1,6391	1,7791	1,6716	0,3811
42	1,7372	1,5663	1,4910	1,7809	1,5504	1,6252	0,2899
43	1,5971	1,7394	1,6832	1,6677	1,7974	1,6970	0,2003
44	1,4295	1,6536	1,9134	1,7272	1,4370	1,6321	0,4839
45	1,6217	1,8220	1,7915	1,6744	1,9404	1,7700	0,3187

4.1 Data Preparation and Phase I Analysis: Establishing Control (In-Control State)

Before we start, we need to prepare our data so that R can use it. We will use the qcc package in R, which is a common tool for control charts.

```
# Data for Flow Width Measurements (microns) for Hard Bake Process - Wafers
# From Montgomery, 8th Ed., pp. 223 (Subgroups 1-25) and 227 (Subgroups 26-45)
# This is our raw data for flow width measurements. Each group of 5 numbers is one
subgroup.
# Subgroup size n = 5
# --- 1: Data Loading and Preparation ---
# Load the qcc package.
library(qcc)


data_raw <- c(
  1.3335, 1.4128, 1.6744, 1.4573, 1.6914,  # Subgroup 1
  1.4314, 1.3592, 1.6075, 1.4666, 1.6109,  # Subgroup 2
  1.4284, 1.4871, 1.4932, 1.4324, 1.5674,  # Subgroup 3
  1.5028, 1.6352, 1.3841, 1.2831, 1.5507,  # Subgroup 4
  1.5604, 1.2735, 1.5265, 1.4363, 1.6441,  # Subgroup 5
  1.5955, 1.5451, 1.3574, 1.3281, 1.4198,  # Subgroup 6
  1.6274, 1.5064, 1.8366, 1.4177, 1.5144,  # Subgroup 7
  1.4190, 1.4303, 1.6637, 1.6067, 1.5519,  # Subgroup 8
```

Then, we will analyze the initial 45 subgroups. This helps us see if the process was historically stable and allows us to set correct control limits for future monitoring.

4.1.1 R Chart for Variability

Following Montgomery's advice, we will first create and analyze the R chart. This chart tells us about the process variability.

```
# Create the R chart (Initial Data) using our data_matrix.
# `type="R"` means we want an R chart.
# We give it a clear title and labels for the axes.
qcc_R_chart <- qcc(data_matrix, type="R", data.name="Flow Width Measurements",
                   title="R Chart for Flow Width (Initial Data)",
                   xlab="Subgroup Number", ylab="Subgroup Range")

# To see the exact control limits and center line values, you can use:
summary(qcc_R_chart)
```

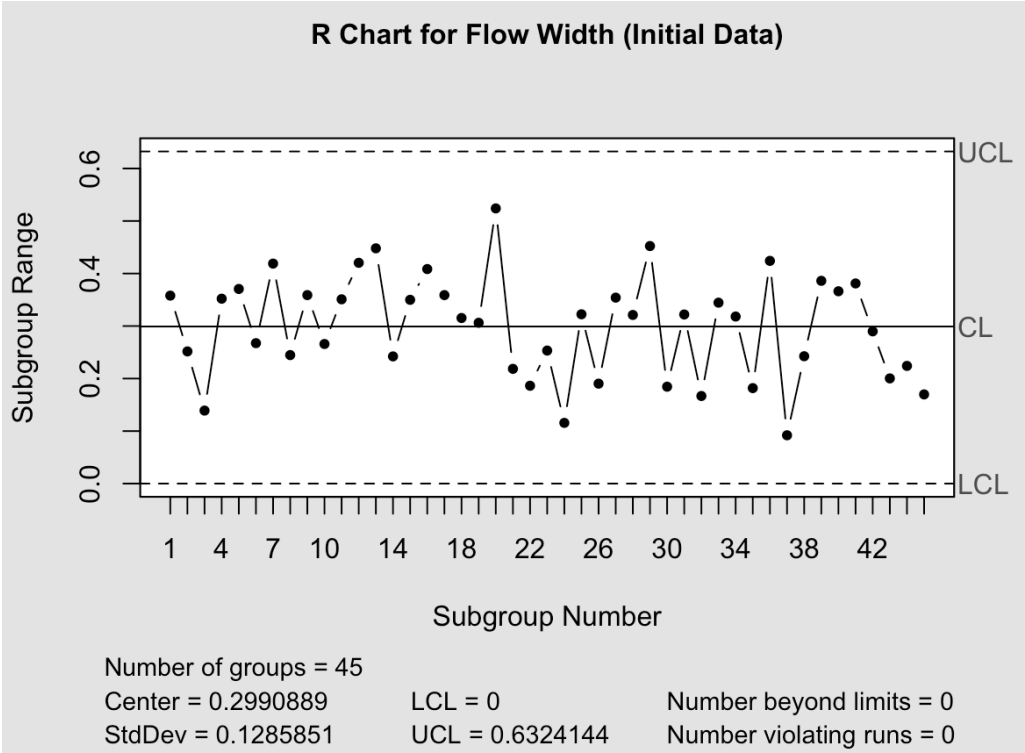


FIGURE 1

Interpretation of the R Chart (Initial Data):

Looking at this R chart (Figure 1), we can see the center line (CL) is at approximately 0.300 microns**.** The upper control limit (UCL) is at about 0.632 microns, and the lower control limit (LCL) is at 0 microns**.**

All the data points on this chart are comfortably **within these control limits**. Also, we do not see any unusual patterns. For example, there are no points beyond limits, no long runs of points on one side of the center line, and no clear trends going up or down. This tells us that **the process variability** was **in statistical control** during the time this data was collected. This is a very good sign because it means the consistency of our wafer flow width was stable. Since the variability is stable, we can now confidently move on to analyze the $\overline{X}$ chart

4.1.2 $\overline{X}$ Chart for the Mean

Now that we know the process variability is in control, we can create and interpret the $\overline{X}$ chart. This chart helps us understand the average of our process

```
# Create the X-bar chart
# We create the X-bar chart using the same data.
# `type="xbar"` means we want an X-bar chart.
# We specify `nsigmas = 3` which is the standard for 3-sigma control limits.

qcc_xbar_chart <- qcc(data_matrix, type="xbar", data.name="Flow Width Measurements",
                      title="X-bar Chart for Flow Width (Initial Data)",
                      xlab="Subgroup Number", ylab="Subgroup Mean",
                      nsigmas = 3) # This sets up the 3-sigma limits automatically

# To see the exact control limits and center line values, you can use:
summary(qcc_xbar_chart)
```

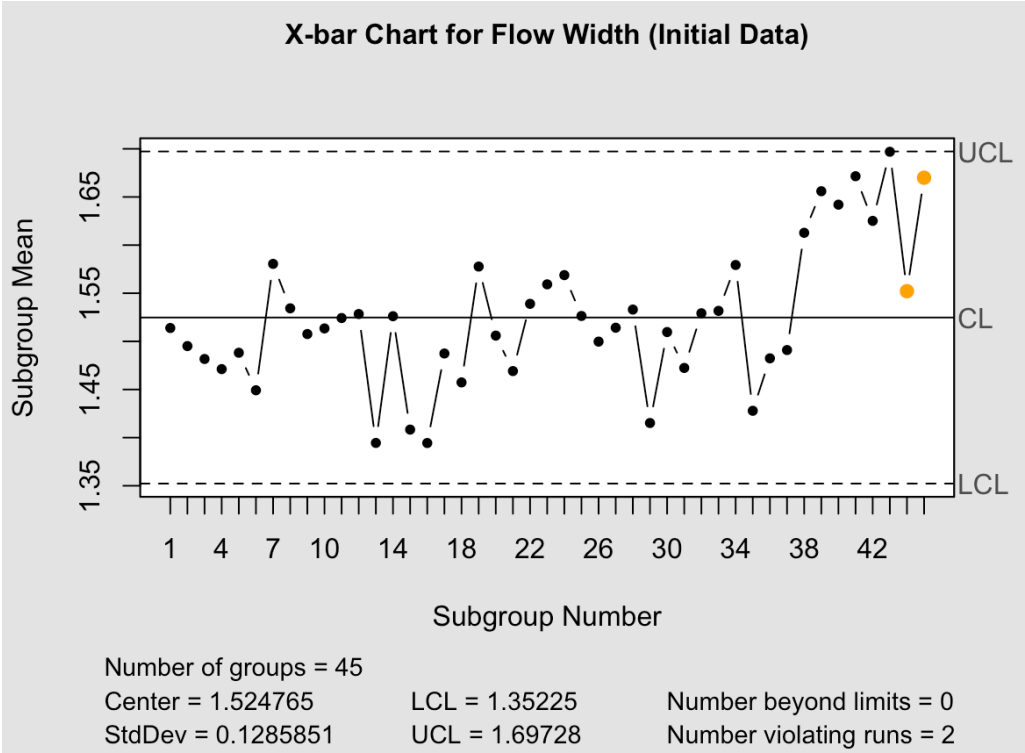


FIGURE 2

Interpretation of the $\bar{X}$ Chart (Initial Data):

This $\bar{X}$ chart (Figure 2) shows us the average flow width for each subgroup over time. The center line (CL) is at approximately 1.524 microns. The upper control limit (UCL) is at about 1.697 microns, and the lower control limit (LCL) is at about **1**.352 microns.

All the subgroup means are **within these control limits**. We also do not observe any unusual patterns, such as points outside limits. This means that **the process mean** was also **in statistical control**.

Joint Conclusion for Initial Data:

Both the R chart and the $\bar{X}$ chart show that this manufacturing process was **stable and predictable**. This means the process was performing consistently in terms of both its average flow width and its variation. This is a good baseline for our process.

4.2 Phase II Analysis : Demonstrating Out-of-Control Scenarios

Now, let's create some common out-of-control situations. This will help us understand how these joint charts work to identify problems quickly. We will change the original data a little bit to simulate these problems. For these new charts, we will use the control limits that we calculated from the initial (in-control) data. This is how we monitor a process in real-time (**Phase II monitoring**).

4.2.1 Scenario 1: A Shift in the Process Mean

Imagine that around subgroup 30, something caused the flow width to increase slightly. For example, a machine setting might have accidentally changed. We will add a small value (0.2 microns) to the measurements from subgroup 30 onwards. This value is chosen to be noticeable given the scale of our current data.

```
# Now, we simulate a shift. From subgroup 30 to subgroup 45, we add 0.2 to each
measurement.
# This makes the mean of these subgroups higher.
data_mean_shift[30:45, ] <- data_mean_shift[30:45, ] + 0.2

# We re-run the R chart first. We expect it to remain in control because a mean shift
# should not affect the process variability. We use the `qcc_R_chart`'s parameters
# to make sure we are monitoring against our established baseline.
qcc_R_shift <- qcc(data_mean_shift, type="R", data.name="Flow Width Measurements (Mean
Shift)",
                   title="R Chart for Flow Width (with Simulated Mean Shift)",
                   xlab="Subgroup Number", ylab="Subgroup Range",
                   center = qcc_R_chart$center,
                   std.dev = qcc_R_chart$std.dev,
                   limits = qcc_R_chart$limits)

# Then, we run the X-bar chart. This chart  show the shift.
# We also use the established parameters from our initial X-bar chart.
qcc_xbar_shift <- qcc(data_mean_shift, type="xbar", data.name="Flow Width Measurements
(Mean Shift)",
                      title="X-bar Chart for Flow Width (with Simulated Mean Shift)",
                      xlab="Subgroup Number", ylab="Subgroup Mean",
                      center = qcc_xbar_chart$center,
                      std.dev = qcc_xbar_chart$std.dev,
                      limits = qcc_xbar_chart$limits)
```

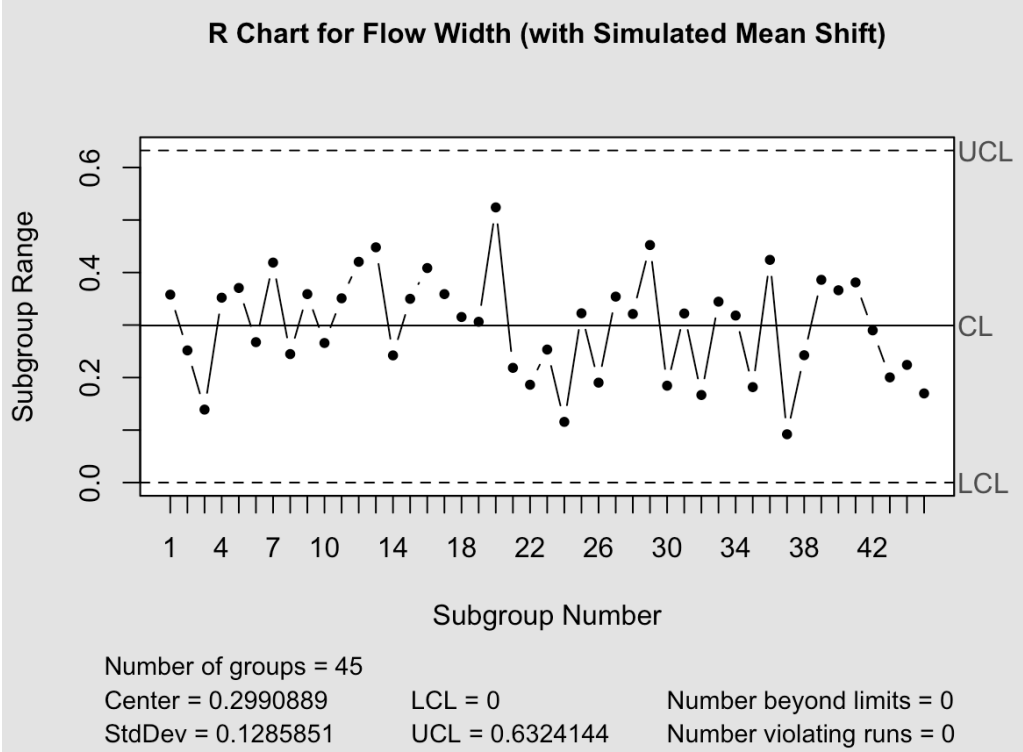


FIGURE 3

Interpretation of R Chart (with Mean Shift):

Looking at this R chart (Figure 3), which shows our simulated mean shift, we still see all points within the control limits. We also don't see any new unusual patterns. This is exactly what we expect! An assignable cause that only shifts the process mean (like a machine setting change) usually does not affect how consistent the process is. So, the process variability remains in control. This means our R chart correctly tells us that the process consistency is fine.

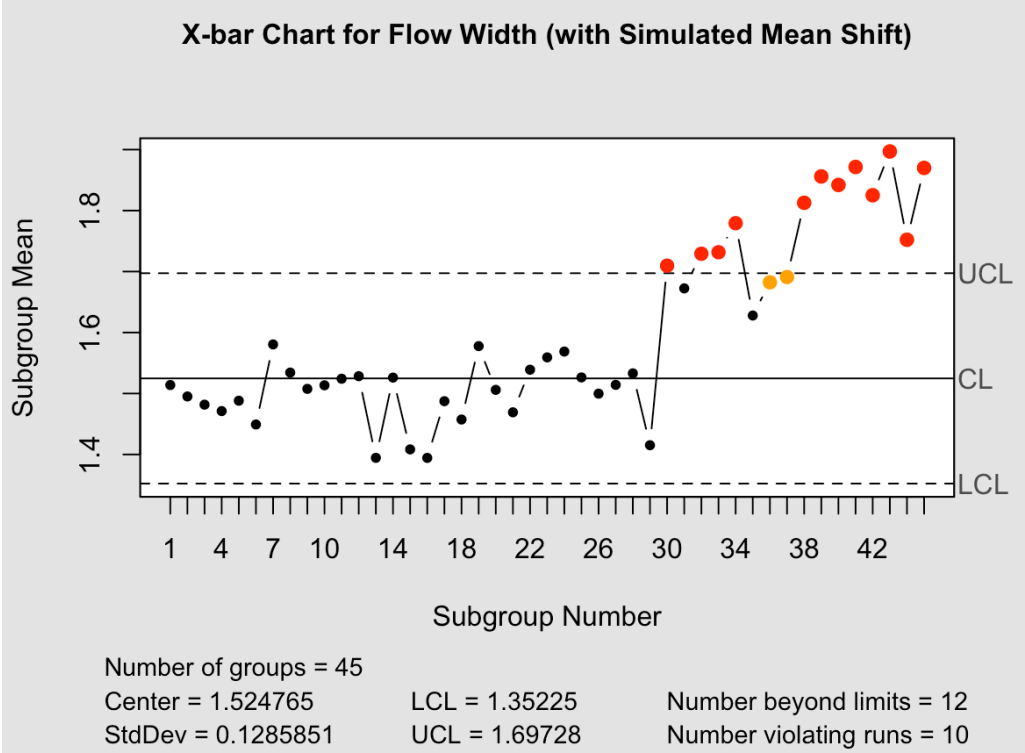


FIGURE 4 - Control limits calculated with α=0.0027 (3-sigma) for each chart.

Interpretation of $\overline{X}$ Chart (with Mean Shift):

Now, let's look at the $\overline{X}$ chart (Figure 4) for this scenario. We can clearly see a big change starting from subgroup 30. Many points from subgroup 30 onwards are now **above the Upper Control Limit (UCL)**. We also see a clear "run" of points all above the center line. This immediate and strong signal tells us that the process mean has shifted upwards. An engineer or operator would need to investigate this immediately. This could be due to something like a machine calibration error or a change in raw material properties that makes the product bigger. This scenario shows how the $\overline{X}$ chart clearly points to a problem with the average

.

4.2.2 Scenario 2: An Increase in Process Variation

Next, let's imagine that around subgroup 30, a tool started to wear out. This caused the flow width measurements to become more spread out (more variable). We will increase the variability by adding some more random differences to the measurements from subgroup 30 onwards. We will use a larger standard deviation for this added noise (e.g., 0.15).

```
##3.2 Scenario 2: An Increase in Process Variation

We simulate an increase in the process variation starting from subgroup 30.

# Make a copy of the original data
data_var_increase <- data_matrix

# Simulate an increase in variation from subgroup 30 onwards.
# We add random noise with a larger standard deviation (e.g., 0.15).
# set.seed() makes sure our random numbers are the same every time we run the code.
set.seed(456)
data_var_increase[30:45, ] <- data_var_increase[30:45, ] + matrix(rnorm(16*5, mean = 0, sd =
0.15), nrow=16, ncol=5)

# We re-run the R chart first. This chart  show the increase in variation.
qcc_R_var_increase <- qcc(data_var_increase, type="R", data.name="Flow Width Measurements (Var
Increase)",
                          title="R Chart for Flow Width (with Simulated Variation Increase)",
                          xlab="Subgroup Number", ylab="Subgroup Range",
                          center = qcc_R_chart$center, # Use initial R chart parameters
                          std.dev = qcc_R_chart$std.dev,
                          limits = qcc_R_chart$limits)

# Then, we run the X-bar chart. It might show some effects, but the R chart is key here.
qcc_xbar_var_increase <- qcc(data_var_increase, type="xbar", data.name="Flow Width Measurements
(Var Increase)",
                             title="X-bar Chart for Flow Width (with Simulated Variation
Increase)",
                             xlab="Subgroup Number", ylab="Subgroup Mean",
                             center = qcc_xbar_chart$center, # Use initial X-bar chart parameters
                             std.dev = qcc_xbar_chart$std.dev,
                             limits = qcc_xbar_chart$limits)
```

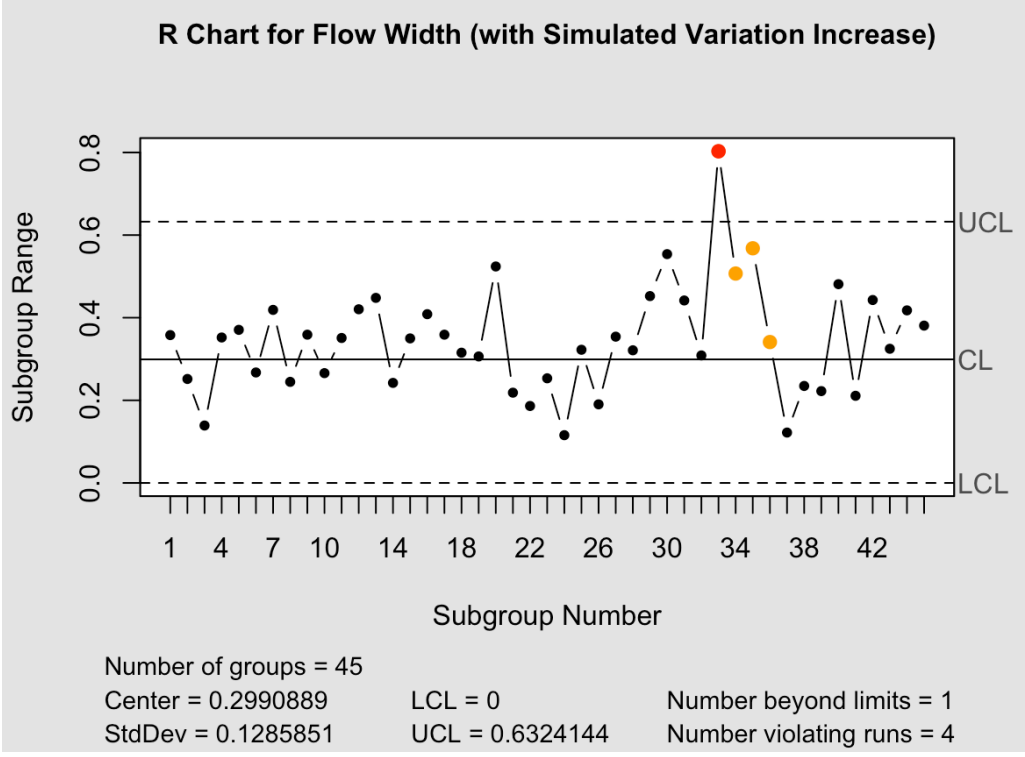


FIGURE 5 - Control limits calculated with α=0.0027 (3-sigma) for each chart.

Interpretation of R Chart (with Variation Increase):

On this R chart (Figure 5), we can clearly see a big change starting from subgroup 30. The points are generally much higher, and several points are now **above the Upper Control Limit (UCL)**. This strongly signals that the process variability has significantly increased. This problem needs immediate investigation! It could be caused by factors like a worn tool, unstable raw material that changes its properties, or operators using inconsistent methods. This chart clearly shows a problem with the consistency of the process.

.

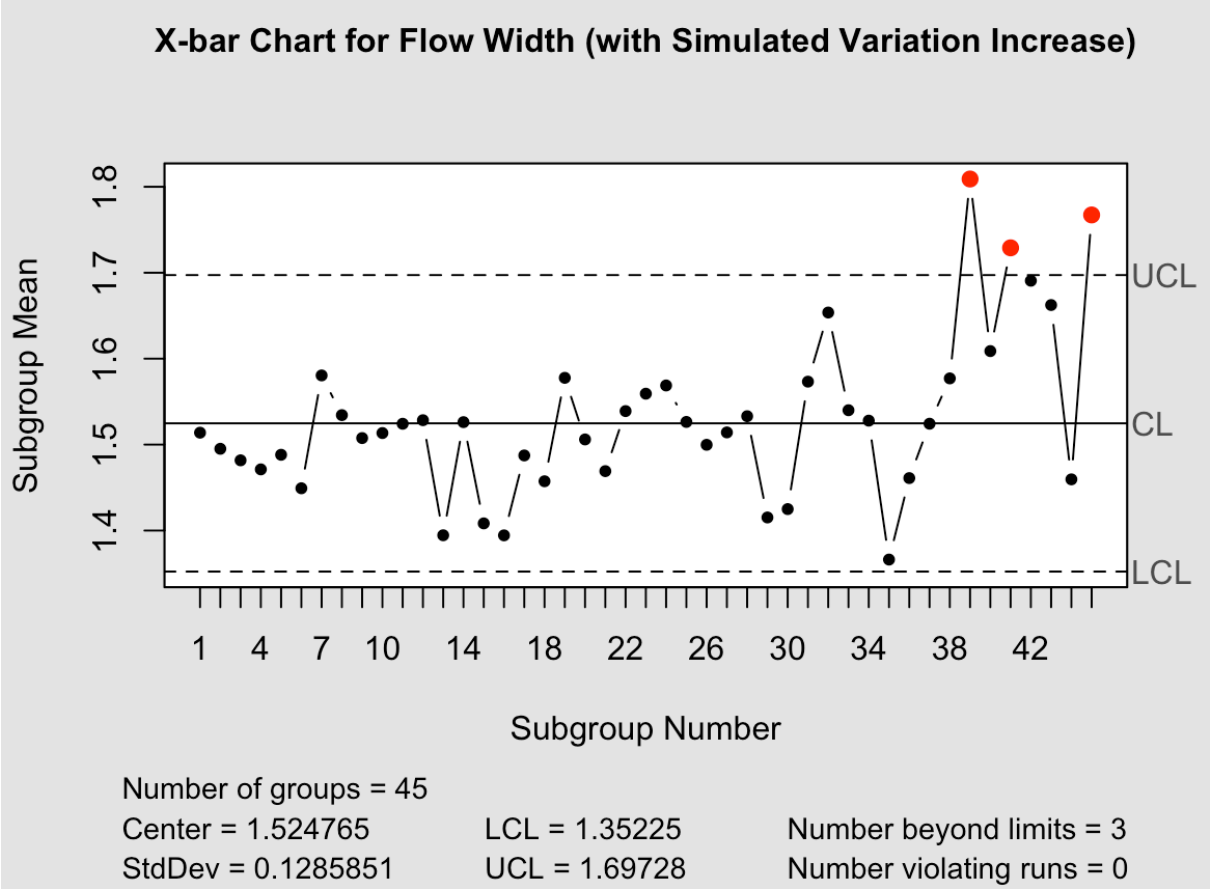


FIGURE 6

Interpretation of $\overline{X}$ Chart (with Variation Increase):

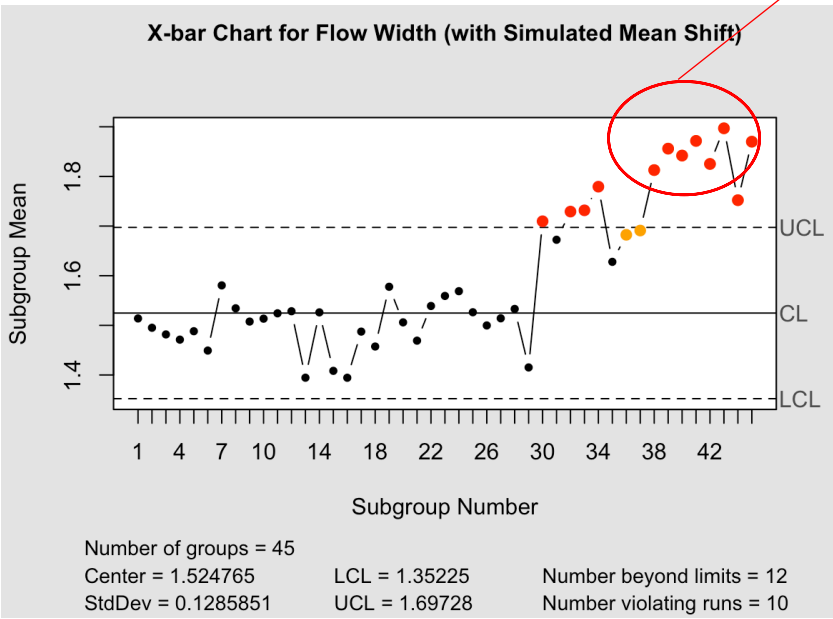
Now, let's look at the $\overline{X}$ chart (Figure 6) for the same scenario. We might see that the points are more spread out, and some points could be closer to or even outside the original limits. This happens because when variability increases, the individual measurements become much wider, which can sometimes push subgroup averages outside their normal range, even if the overall process mean has not actually changed. However, the **strongest and most direct signal** for this type of problem comes from the **R chart**. This example clearly shows why Montgomery's rule is so important: **always check the variability chart first!** If the R chart shows a problem, the $\overline{X}$ chart might be misleading, or its signals might be secondary effects. We must fix the variation problem before truly trusting the $\overline{X}$ chart.

4.3 Interpreting Control Chart Signals: What to Look For

Control charts don't just detect if a point goes beyond the limits. They also look for specific patterns that signal an "assignable cause" , a special problem in the process. Recognizing these patterns is crucial for quick intervention. The qcc package automatically flags some of these key signals.

1. **Points Beyond Control Limits (Red Dots ●);**

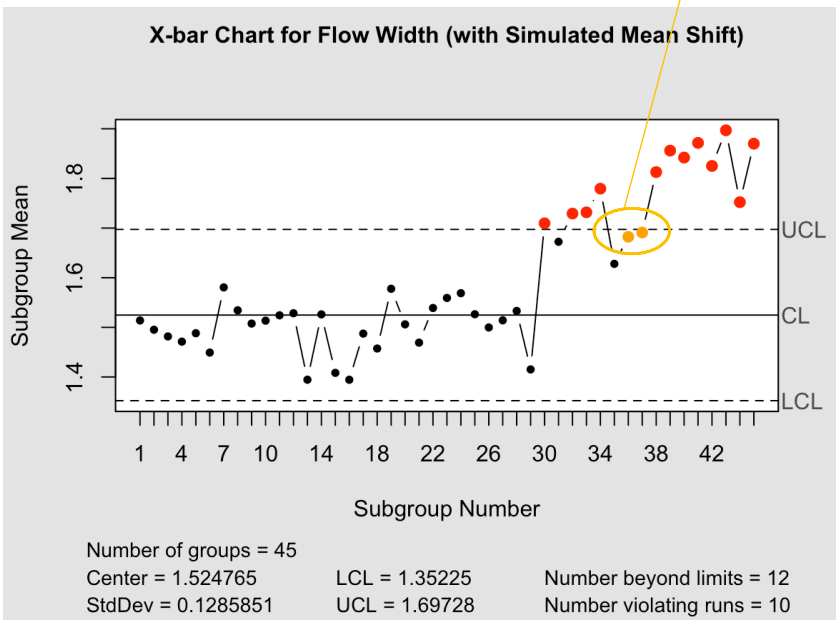
- **What it means:** This is the most obvious and critical signal. If a data point (subgroup mean on X-bar chart or range on R chart) falls outside the Upper Control Limit (UCL) or Lower Control Limit (LCL), it means the process is no longer operating within its usual, predictable boundaries.
- **Why it's important:** It strongly indicates a major, immediate problem that requires urgent investigation.
- **Example from our simulations:** As seen in our X-bar Chart with Simulated Mean Shift (Figure 4), many red points appeared above the UCL starting from subgroup 30. This immediately told us that the process average had significantly increased.
-



2. **Runs (Consecutive Points on One Side of the Center Line) (Orange Dots ●) :**

- **What it means:** When a certain number of consecutive points (e.g., 7 or 8 points, depending on the specific rule) fall on the same side of the center line (all above or all below), it suggests a subtle but persistent shift in the process. The orange dots on our charts indicate points that are part of such a run.
- **Why it's important:** This is an earlier warning sign for a potential shift that might not yet have crossed the control limits. It can indicate a small, gradual change.
- **Example from our simulations:** In our X-bar Chart with Simulated Mean Shift (Figure 4), even before points crossed the UCL, we might notice patterns of orange dots showing

consecutive points above the center line. This would have been an early indicator of the upward shift , an early warning sign .

- 

3. **Trends (Consecutive Points Trending Up or Down):**
    - **What it means:** A series of consecutive points (e.g., 6 or more) that are continuously increasing or decreasing indicates a steady, systematic change in the process over time.
    - **Why it's important:** This suggests a gradual deterioration or improvement that needs to be understood. It could be tool wear, material build-up, or gradual temperature changes.
    - **Example:** "While our specific simulations focused on sudden shifts, a trend could look like points gradually moving upwards or downwards on the chart." (Senin grafiklerinde bariz bir trend yoktu, bu yüzden genel bir açıklama yeterli.)
2. **Periodicity or Cycles:**
    - **What it means:** If points show a repeating up-and-down pattern, it suggests a cyclical cause like changes in operators, shifts, or environmental conditions.
    - **Why it's important:** Helps identify problems tied to specific repeating events.

Conclusion for Signal Interpretation: By understanding these different types of signals, process engineers can quickly diagnose and react to potential issues. It's not just about points outside limits; the ***pattern*** of the points tells a story about the process.

5. Conclusion

Joint $\overline{X}$ and R/S control charts are truly fundamental tools in Statistical Process Control, providing a powerful and comprehensive approach to process monitoring. They allow us to simultaneously evaluate both a process's average performance and its consistency, addressing the crucial advice from Montgomery to **always examine variability first** to ensure the validity of mean chart interpretations.

Through our detailed analysis of the flow width data, we have clearly seen how these two charts complement each other. The R chart effectively detected the simulated changes in process spread, while the $\overline{X}$ chart immediately signaled shifts in the process average. By combining their insights, we gain a holistic understanding of the process state.

By consistently applying these charts and acting promptly on out-of-control signals – whether they are points beyond limits, violating runs, or other patterns – we can rapidly identify and eliminate assignable causes of variation. This proactive approach leads directly to more stable and predictable processes, significantly reducing defects, minimizing rework, and ultimately lowering costs. The cumulative effect is higher quality products, increased customer satisfaction, and a stronger competitive position. Therefore, control charts are not merely problem detectors; they are indispensable instruments for driving continuous process improvement and maintaining long-term process capability.

6. References

- Montgomery, D. C. (2021). "Introduction to Statistical Quality Control *(8th ed.*)". John Wiley & Sons.
- Grant, E. L. (1952). "Statistical Quality Control 2nd ed." McGraw-Hill.
- Bissell, D. (1994). "Statistical Methods for SPC and TQM. Chapman & Hall/CRC Texts in Statistical Science,Vol. 26." . Chapman & Hall London.

7. Appendix

7.1 Factors For Constructing Variables Charts

A-12 **Appendix VI** Factors for Constructing Variables Control Charts

	Chart for Averages					Chart for Standard Deviations						Chart for Ranges				
	Factors for Control Limits			Factors for Center Line		Factors for Control Limits				Factors for Center Line		Factors for Control Limits				
Observations in Sample, n	A	A_2	A_3	c_4	$1/c_4$	B_3	B_4	B_5	B_6	d_2	$1/d_2$	d_3	D_1	D_2	D_3	D_4
2	2.121	1.880	2.659	0.7979	1.2533	0	3.267	0	2.606	1.128	0.8865	0.853	0	3.686	0	3.267
3	1.732	1.023	1.954	0.8862	1.1284	0	2.568	0	2.276	1.693	0.5907	0.888	0	4.358	0	2.574
4	1.500	0.729	1.628	0.9213	1.0854	0	2.266	0	2.088	2.059	0.4857	0.880	0	4.698	0	2.282
5	1.342	0.577	1.427	0.9400	1.0638	0	2.089	0	1.964	2.326	0.4299	0.864	0	4.918	0	2.114
6	1.225	0.483	1.287	0.9515	1.0510	0.030	1.970	0.029	1.874	2.534	0.3946	0.848	0	5.078	0	2.004
7	1.134	0.419	1.182	0.9594	1.0423	0.118	1.882	0.113	1.806	2.704	0.3698	0.833	0.204	5.204	0.076	1.924
8	1.061	0.373	1.099	0.9650	1.0363	0.185	1.815	0.179	1.751	2.847	0.3512	0.820	0.388	5.306	0.136	1.864
9	1.000	0.337	1.032	0.9693	1.0317	0.239	1.761	0.232	1.707	2.970	0.3367	0.808	0.547	5.393	0.184	1.816
10	0.949	0.308	0.975	0.9727	1.0281	0.284	1.716	0.276	1.669	3.078	0.3249	0.797	0.687	5.469	0.223	1.777
11	0.905	0.285	0.927	0.9754	1.0252	0.321	1.679	0.313	1.637	3.173	0.3152	0.787	0.811	5.535	0.256	1.744
12	0.866	0.266	0.886	0.9776	1.0229	0.354	1.646	0.346	1.610	3.258	0.3069	0.778	0.922	5.594	0.283	1.717
13	0.832	0.249	0.850	0.9794	1.0210	0.382	1.618	0.374	1.585	3.336	0.2998	0.770	1.025	5.647	0.307	1.693
14	0.802	0.235	0.817	0.9810	1.0194	0.406	1.594	0.399	1.563	3.407	0.2935	0.763	1.118	5.696	0.328	1.672
15	0.775	0.223	0.789	0.9823	1.0180	0.428	1.572	0.421	1.544	3.472	0.2880	0.756	1.203	5.741	0.347	1.653
16	0.750	0.212	0.763	0.9835	1.0168	0.448	1.552	0.440	1.526	3.532	0.2831	0.750	1.282	5.782	0.363	1.637
17	0.728	0.203	0.739	0.9845	1.0157	0.466	1.534	0.458	1.511	3.588	0.2787	0.744	1.356	5.820	0.378	1.622
18	0.707	0.194	0.718	0.9854	1.0148	0.482	1.518	0.475	1.496	3.640	0.2747	0.739	1.424	5.856	0.391	1.608
19	0.688	0.187	0.698	0.9862	1.0140	0.497	1.503	0.490	1.483	3.689	0.2711	0.734	1.487	5.891	0.403	1.597
20	0.671	0.180	0.680	0.9869	1.0133	0.510	1.490	0.504	1.470	3.735	0.2677	0.729	1.549	5.921	0.415	1.585
21	0.655	0.173	0.663	0.9876	1.0126	0.523	1.477	0.516	1.459	3.778	0.2647	0.724	1.605	5.951	0.425	1.575
22	0.640	0.167	0.647	0.9882	1.0119	0.534	1.466	0.528	1.448	3.819	0.2618	0.720	1.659	5.979	0.434	1.566
23	0.626	0.162	0.633	0.9887	1.0114	0.545	1.455	0.539	1.438	3.858	0.2592	0.716	1.710	6.006	0.443	1.557
24	0.612	0.157	0.619	0.9892	1.0109	0.555	1.445	0.549	1.429	3.895	0.2567	0.712	1.759	6.031	0.451	1.548
25	0.600	0.153	0.606	0.9896	1.0105	0.565	1.435	0.559	1.420	3.931	0.2544	0.708	1.806	6.056	0.459	1.541

For $n > 25$.